

Manipulating Data

Data Manipulation Language (DML)

A DML statement is executed when you:

- Add new rows to a table
- Modify existing rows in a table
- Remove existing rows from a table

Adding a New Row to a Table

DEPARTMENTS

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
50	Shipping	124	1500
60	IT	103	1400
80	Sales	149	2500
90	Executive	100	1700
110	Accounting	205	1700
190	Contracting		1700

70	Public Relations	100	1700
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**New
row**

**Insert a new row
into the
DEPARMENTS table.**



DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
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20	Marketing	201	1800
50	Shipping	124	1500
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90	Executive	100	1700
110	Accounting	205	1700
190	Contracting		1700
70	Public Relations	100	1700

The INSERT Statement Syntax

- Add new rows to a table by using the INSERT statement

```
INSERT INTO  table [(column [, column...])]  
VALUES      (value [, value...]);
```

- Only one row is inserted at a time with this syntax

The INSERT Statement Syntax(Contd..)

In the syntax:

- **table**
is the name of the table
- **column**
is the name of the column in the table to populate
- **values**
is the corresponding value for the column

Inserting New Rows

- Insert a new row containing values for each column
- List values in the default order of the columns in the table

```
INSERT INTO departments(department_id, department_name,  
                        manager_id, location_id)  
VALUES      (70, 'Public Relations', 100, 1700);  
1 row created.
```

- Enclose character and date values within single quotation marks

Inserting Rows with Null Values

DESCRIBE departments

Name	Null?	Type
DEPARTMENT_ID	NOT NULL	NUMBER(4)
DEPARTMENT_NAME	NOT NULL	VARCHAR2(30)
MANAGER_ID		NUMBER(6)
LOCATION_ID		NUMBER(4)

- Implicit method: Omit the column from the column list

```
INSERT INTO departments (department_id,  
                        department_name )  
VALUES (30, 'Purchasing');  
1 row created.
```


Inserting Rows with Null Values(Contd..)

- Explicit method: Specify the NULL keyword in the VALUES clause

```
INSERT INTO departments
VALUES      (100, 'Finance', NULL, NULL);
1 row created.
```


Inserting Special Values

- The SYSDATE function records the current date and time.

```
INSERT INTO employees (employee_id,  
                        first_name, last_name,  
                        email, phone_number,  
                        hire_date, job_id, salary,  
                        commission_pct, manager_id,  
                        department_id)  
VALUES  
      (113,  
       'Louis', 'Popp',  
       'LPOPP', '515.124.4567',  
       SYSDATE, 'AC_ACCOUNT', 6900,  
       NULL, 205, 100);
```

```
1 row created.
```

Inserting Specific Date Values

- Add a new employee

```
INSERT INTO employees
VALUES      (114,
             'Den', 'Raphealy',
             'DRAPHEAL', '515.127.4561',
             TO_DATE('FEB 3, 1999', 'MON DD, YYYY'),
             'AC_ACCOUNT', 11000, NULL, 100, 30);

1 row created.
```

Insert values into the table

- Use & substitution in a SQL statement to prompt for values
- & is a placeholder for the variable value

```
INSERT INTO departments
        (department_id, department_name, location_id)
VALUES    (&department_id, '&department_name', &location);
```

Define Substitution Variables

department_id

department_name

location

```
1 row created.
```


Changing Data in a Table

EMPLOYEES

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	HIRE_DATE	JOB_ID	SALARY	DEPARTMENT_ID	COMM.
100	Steven	King	SKING	17-JUN-87	AD_PRES	24000	90	
101	Neena	Kochhar	NKOCHHAR	21-SEP-89	AD_VP	17000	90	
102	Lex	De Haan	LDEHAAN	13-JAN-93	AD_VP	17000	90	
103	Alexander	Hunold	AHUNOLD	03-JAN-90	IT_PROG	9000	60	
104	Bruce	Ernst	BERNST	21-MAY-91	IT_PROG	6000	60	
107	Diana	Lorentz	DLORENTZ	07-FEB-99	IT_PROG	4200	60	
124	Kevin	Mourgos	KMOURGOS	16-NOV-99	ST_MAN	5800	50	

Update rows in the **EMPLOYEES** table.



EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	HIRE_DATE	JOB_ID	SALARY	DEPARTMENT_ID	COMM.
100	Steven	King	SKING	17-JUN-87	AD_PRES	24000	90	
101	Neena	Kochhar	NKOCHHAR	21-SEP-89	AD_VP	17000	90	
102	Lex	De Haan	LDEHAAN	13-JAN-93	AD_VP	17000	90	
103	Alexander	Hunold	AHUNOLD	03-JAN-90	IT_PROG	9000	60	
104	Bruce	Ernst	BERNST	21-MAY-91	IT_PROG	6000	60	
107	Diana	Lorentz	DLORENTZ	07-FEB-99	IT_PROG	4200	60	
124	Kevin	Mourgos	KMOURGOS	16-NOV-99	ST_MAN	5800	50	

The UPDATE Statement Syntax

- Modify existing rows with the UPDATE statement

```
UPDATE      table
SET         column = value [, column = value, ...]
[WHERE     condition];
```

- Update more than one row at a time, if required

Updating Rows in a Table

- Specific row or rows are modified if you specify the WHERE clause

```
UPDATE employees  
SET    department_id = 70  
WHERE  employee_id = 113;  
1 row updated.
```

- All rows in the table are modified if you omit the WHERE clause

```
UPDATE    copy_emp  
SET       department_id = 110;  
22 rows updated.
```

Removing a Row from a Table

DEPARTMENTS

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
70	Public Relations	100	1700
30	Purchasing		
50	Shipping	124	1500
60	IT	103	1400
100	Finance		
80	Sales	149	2500

Delete a row from the DEPARTMENTS table.

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
70	Public Relations	100	1700
30	Purchasing		
50	Shipping	124	1500
60	IT	103	1400
80	Sales	149	2500

The DELETE Statement

- You can remove existing rows from a table by using the DELETE statement

```
DELETE [FROM] table  
[WHERE condition];
```


Deleting Rows from a Table

- Specific rows are deleted if you specify the WHERE clause

```
DELETE FROM departments  
WHERE department_name = 'Finance';  
1 row deleted.
```

- All rows in the table are deleted if you omit the WHERE clause

```
DELETE FROM copy_emp;  
22 rows deleted.
```

Q/A